A Modular Classical Reasoning Agent for RuleTaker, CounterBench, and FOLIO

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***Abstract*—This is a modular classical reasoning agent that routes the various structured reasoning tasks to the relevant benchmark-specific components. RuleTaker is dealt with via forward and backward chaining. CounterBench is dealt with via separate causal reasoning implementations, which include a frozen polarity based causal inference (PBCI) component for Alpha V1.0 and a Backdoor V2.1 version. FOLIO is dealt with via first order logic parsing, an abstract syntax tree and Z3 based entailment checking. On the reported public splits RuleTaker achieves 97.60% test accuracy and 99.47% OOD-test accuracy. The frozen CounterBench Alpha PBCI V1.0 component achieves 90.00% on 1,000 examples and Backdoor V2.1 achieves 90.00% on 200 examples. FOLIO achieves 81.32% training accuracy and 84.73% validation accuracy. FOLIO parser coverage is 89.51% on training and 94.58% on validation and parser clean accuracies are 83.15% and 85.94% respectively. The results are reported separately as the benchmarks and splits are measuring different reasoning tasks.**

***Index Terms*—classical reasoning, RuleTaker, CounterBench, causal inference, first-order logic, Z3, entailment**

I. Introduction

Reliable reasoning over explicit facts, rules, causal mechanisms and quantified logical statements is still a very difficult problem. The benchmarks considered here use different input representations and reasoning tasks. RuleTaker is for rule-based entailment, CounterBench is for causal and counterfactual reasoning and FOLIO is for first order logic premises and conclusions.

A single shared inference procedure is not appropriate for all of these representations. The system implemented therefore adopts a task-routed architecture in which a common interface selects a specialised parser and reasoner, with each component retaining its own internal representation and inference procedure. The aim is integration and transparent evaluation, not the training of a single unified neural model.

II. System Architecture

The agent consists of the following components:

1) RuleTaker natural-language fact/rule parsing and forward/backward chaining.
2) CounterBench Alpha PBCI V1.0 frozen reference component.
3) CounterBench Backdoor V2.1 causal reasoning adaptation.
4) FOLIO first-order formula parsing and Z3 entailment checking.

Each branch returns a Boolean prediction where applicable. The task dispatcher identifies the relevant branch from the task field or supported input-column pairs.

A. RuleTaker

The RuleTaker parser takes context statements and turns them into facts and implication rules. Forward chaining is the process of repeatedly applying rules whose antecedents are satisfied until no more facts can be derived. Backward chaining is the process of starting from the query and then recursively attempting to establish the antecedents of the query. The notebook performs an evaluation of both of these methods on the training, test and OOD-test splits that are available. Both methods achieve the same reported accuracy on these splits.

B. CounterBench Alpha PBCI V1.0

The Alpha PBCI V1.0 implementation is kept as a frozen reference component. The final notebook loads the Alpha dataset for reference and deliberately does not re-run or re-apply the frozen evaluator. Its recorded result is reported separately: 900 correct out of 1,000 (90.00%). The recorded precision is 87.04%, the recall is 94.00% and the F1-score is 90.38%. These are the results of the frozen Alpha PBCI V1.0 evaluation and should not be confused with the separate general causal mechanism parser/simulator experiment which was part of earlier development.

C. CounterBench Backdoor V2.1

Backdoor V2.1 employs a separate causal reasoning procedure. The adapted workflow incorporates root-world reasoning, intervention and prediction. It's been evaluated on 200 labelled examples. The recorded confusion matrix, with rows as actual labels and columns as predicted labels in the order False/True, is

$$\begin{bmatrix} 100 & 5 \\ 15 & 80 \end{bmatrix}.$$

The system obtains 90.00% accuracy, 94.12% precision, 84.21% recall, and 88.89% F1-score. The 95% Wilson confidence interval for accuracy is [85.06%, 93.43%]. This is a labelled public evaluation result, not a private-test result.

D. FOLIO First-Order Entailment

FOLIO formulae are tokenised and parsed into an abstract syntax tree. The implementation supports universal and existential quantifiers, negation, conjunction, disjunction, implication, biconditional, exclusive OR, predicates and equality (subject to the parser grammar and the term representation that is supported).

The AST is translated into Z3 formulas over an uninterpreted domain. Entailment is checked by testing whether the premises together with the negated conclusion are unsatisfiable:

$$KB \models \phi \quad \Longleftrightarrow \quad KB \wedge \neg\phi \text{ is UNSAT},$$

provided the premises are consistent. The solver uses an 8,000 ms timeout and conservative fuzzy vocabulary alignment. FOLIO labels True as positive entailment and maps False/Uncertain to the negative class for binary evaluation.

A conservative parser recovery wrapper handles a limited case in which one to three closing parentheses are missing at end-of-input, provided the strict parser reports the expected end-of-input error and the formula can be parsed after appending the missing closers. Other parser/translation failures are simply explicit failures. An additional experimental repair for the case of extra terminal closing parentheses was looked at but not incorporated into the final implementation.

III. Experimental Setup

All the values reported are taken from the outputs of the executed assignment notebooks. RuleTaker uses 1k training, 250 test and 375 OOD test examples. CounterBench Alpha PBCI V1.0 uses 1k examples. CounterBench Backdoor V2.1 uses 200 examples. FOLIO uses 1k1 training and 203 validation examples.

Accuracy is the fraction of correct predictions. Precision, recall and F1-score (for the positive class where available) are reported. For FOLIO the overall metrics are on all examples and parser-clean metrics are on the subset of examples with no parse/translation errors. Parser coverage is reported along with parser-clean accuracy to make the evaluated subset explicit.

IV. Results

A. RuleTaker

TABLE I
RuleTaker results

Split	Examples	Correct	Accuracy
Train	1,000	983	98.30%
Test	250	244	97.60%
OOD test	375	373	99.47%

Forward and backward chaining produced identical reported accuracy on all three splits. The test and OOD-test confusion matrices were respectively

$$\begin{bmatrix} 88 & 6 \\ 0 & 156 \end{bmatrix}, \qquad \begin{bmatrix} 242 & 2 \\ 0 & 131 \end{bmatrix},$$

with rows and columns ordered False, True.

B. CounterBench

TABLE II
CounterBench results

Component	N	Accuracy	Precision	Recall
Alpha PBCI V1.0	1,000	90.00%	87.04%	94.00%
Backdoor V2.1	200	90.00%	94.12%	84.21%

Backdoor V2.1 has F1-score 88.89% and a 95% Wilson accuracy interval of [85.06%, 93.43%]. Alpha PBCI V1.0 has F1-score 90.38%. These are distinct CounterBench components and their scores are not combined.

C. FOLIO

TABLE III
FOLIO overall and parser-clean results

Metric	Train	Validation
Examples	1,001	203
Correct (all examples)	814	172
Overall accuracy	81.32%	84.73%
Precision (all examples)	90.36%	90.20%
Recall (all examples)	57.99%	63.89%
F1 (all examples)	70.64%	74.80%
Parser coverage	89.51%	94.58%
Parser failures	105	11
Parser-clean accuracy	83.15%	85.94%
Parser-clean precision	90.36%	90.20%
Parser-clean recall	63.92%	67.65%
Parser-clean F1	74.88%	77.31%

The all-example confusion matrices are

$$\begin{bmatrix} 589 & 24 \\ 163 & 225 \end{bmatrix} \text{ (train)}, \qquad \begin{bmatrix} 126 & 5 \\ 26 & 46 \end{bmatrix} \text{ (validation)}.$$

Training parser coverage is 896/1001 and validation coverage is 192/203. Parser-clean accuracy is predicated on successful parsing and translation so it has to be viewed in conjunction with coverage rather than as an alternative to overall accuracy.

D. Combined Public Prediction Archive

The final public eval archive has 3,029 predictions across RuleTaker train/test/OOD-test, CounterBench Backdoor V2.1 public labelled eval and FOLIO train/validation. 2,766 of those are correct, giving a descriptive aggregate of 91.32%. This is an aggregate across datasets and split types and is not a normalised benchmark score. It is included purely as an operational summary. CounterBench Alpha PBCI V1.0 is reported separately and is not part of this 3029 row aggregate.

The current `submission.csv` is a public fallback with 1,028 held-out public predictions: RuleTaker test and OOD-test (625), CounterBench Backdoor V2.1 (200), and FOLIO validation (203). When the instructor provides private test data, the private inference cell is intended to replace this file with predictions for the private records. No private-test accuracy is claimed.

V. Error Analysis and Limitations

The results illustrate that representation coverage and reasoning accuracy are different. For FOLIO, 105 training and 11 validation examples could not be parsed or translated by the final pipeline. Parser-clean metrics describe only the successfully processed subset. Inconsistent knowledge bases and solver-unknown cases are also distinct statuses and should not be conflated with parser failures.

The FOLIO parser is a bounded grammar with conservative recovery and is not in any sense a complete general purpose first order logic front end. Fuzzy vocabulary alignment is a heuristic and it is possible for it to incorrectly align two different terms. Z3 timeouts and unknowns are mapped to a negative prediction in the binary interface. The quantified formulas are expressed using uninterpreted predicates and an uninterpreted non-empty domain so the actual behaviour is very much dependent on the encoding and solver configuration.

CounterBench Alpha PBCI V1.0 is frozen and its score is a benchmark result for that implementation. Backdoor V2.1 is a separate adaptation. Neither score in itself validates all the causal assumptions or guarantees generalisation beyond the examples that have been evaluated. RuleTaker scores are contingent upon the supported context/rule syntax and the benchmark splits.

Finally the 91.32% combined figure is a mixture of training, test, OOD-test and validation examples and should not be used as a single comparative benchmark figure. The individual benchmark figures are the appropriate basis for interpretation.

VI. Conclusion

The project brings together a number of the classic reasoning procedures under a common task routed interface. RuleTaker performs very well on both the reported test and OOD-test splits. The two CounterBench components each achieve 90.00% on their respective public benchmarks and are clearly distinct implementations. FOLIO achieves 84.73% overall validation accuracy and 85.94% parser-clean validation accuracy at 94.58% parser coverage. The report makes a clear distinction between full-split and parser-clean accuracy and does document the remaining parsing limitations. The current public prediction file is a fallback and will be superseded by private-test inference once the private dataset is made available.